

LECTURE 10

Responsive Design and Frameworks

Building one site that adapts to any screen — and the two major CSS frameworks that make it faster.

In This Lecture

- 1 Mobile-first design, the viewport meta tag, and fluid units
- 2 Media queries and breakpoints — every combination, plus responsive images
- 3 Component-based (Bootstrap) vs. utility-first (Tailwind) frameworks
- 4 Bootstrap's grid system in depth: auto-layout, nesting, offsets, ordering
- 5 Tailwind's utility classes, config file, and dark mode
- 6 Extending either framework with your own custom classes
- 7 CSS preprocessors: Sass, SCSS, and LESS — and customizing Bootstrap with Sass

Mobile-First Design

- 1 Write base CSS for small screens FIRST, then add styling for larger screens with media queries
- 2 The opposite of "desktop-first," where you design big and cram it onto a phone afterward
- 3 Mobile traffic is the majority of web visits — adding complexity as space grows is easier than stripping it away

The Viewport Meta Tag

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

width=device-width uses the real device width; initial-scale=1.0 starts at 100% zoom.

Never Forget This Tag

WARNING

Without the viewport meta tag, none of your media queries will work correctly on a real phone — the browser will still pretend it has a much wider screen.

Units That Scale, Not Fixed Pixels

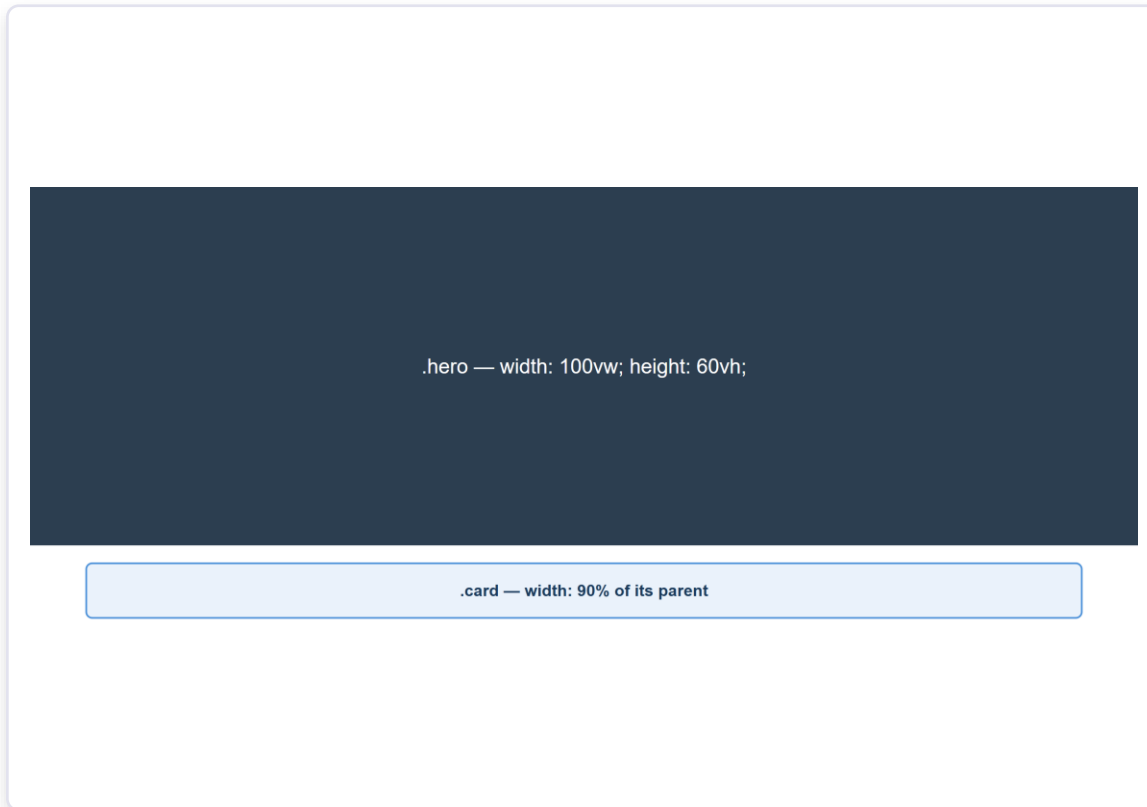
Unit	Relative To
%	The size of the parent element
rem	The root (<html>) element's font size
em	The current element's own font size
vw / vh	1% of the viewport's width / height

Fluid Units in Practice

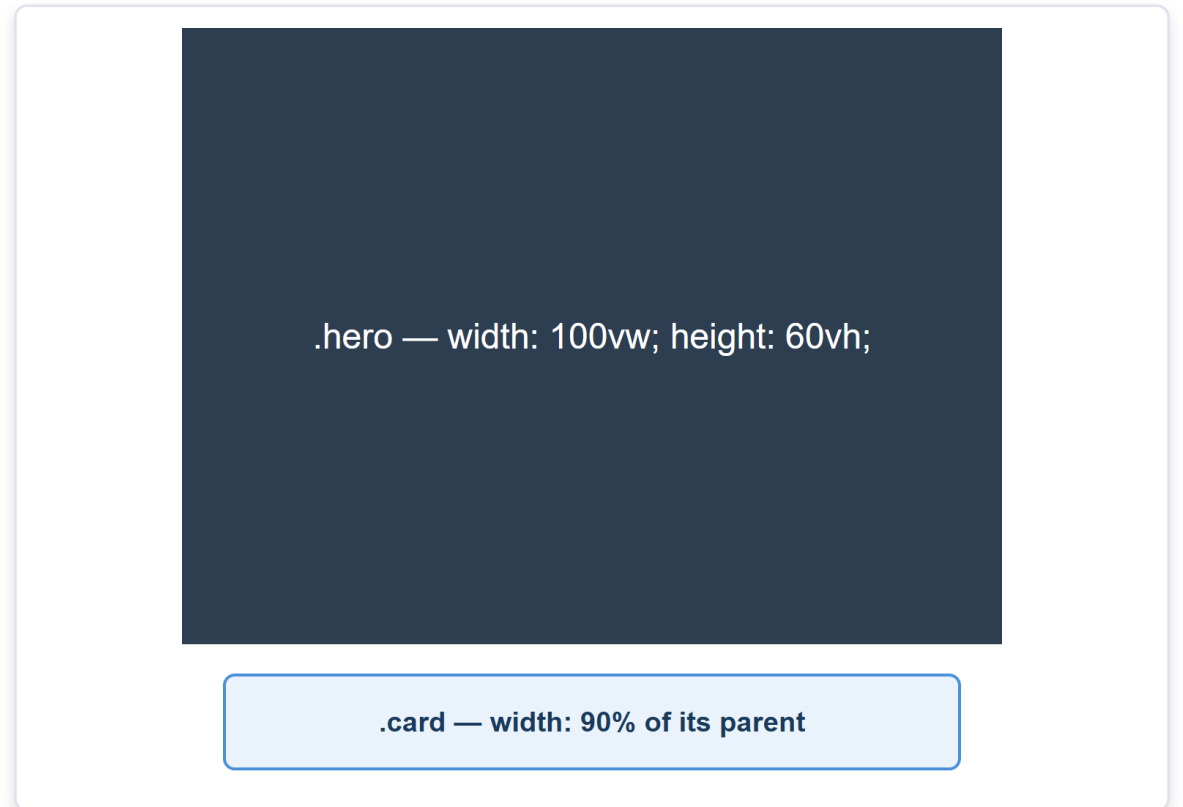
```
.hero {  
  width: 100vw;    /* always fills the full screen width */  
  height: 60vh;    /* always 60% of the visible screen height */  
}  
.card { width: 90%; } /* always 90% of its parent, whatever that is */
```

The Same Page, Two Widths

WIDE



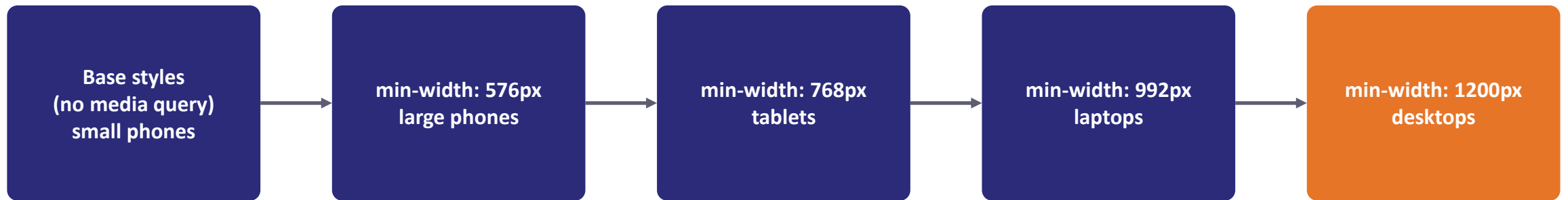
NARROW



A Mobile-First Breakpoint

```
.container { display: flex; flex-direction: column; }  
  
@media (min-width: 768px) {  
  .container { flex-direction: row; }  
}
```

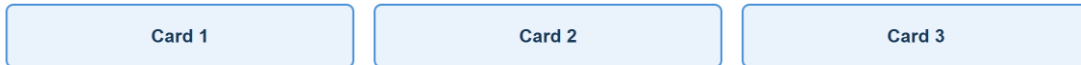
Mobile-First Breakpoint Ladder



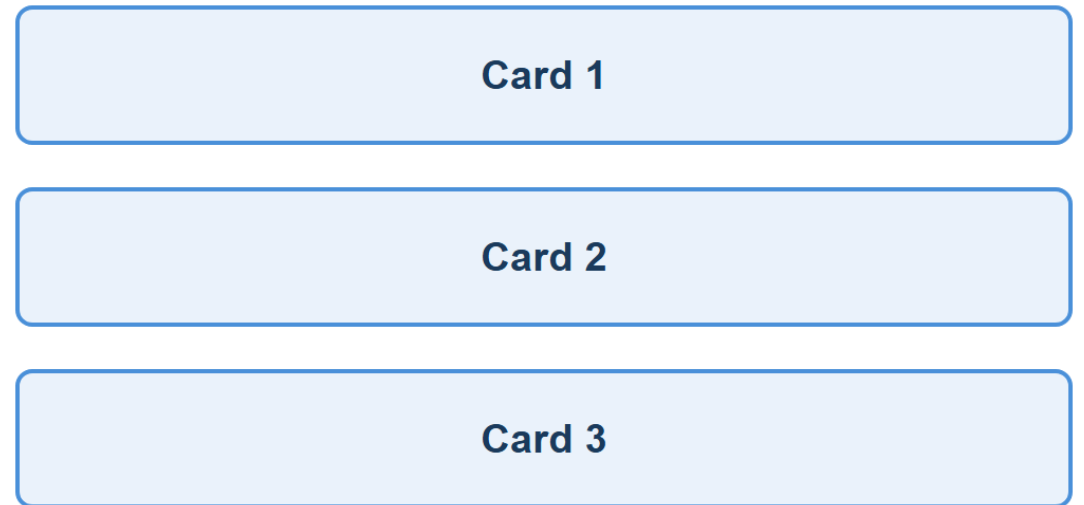
Mobile-first means the base (unqueried) styles target the smallest screen — each breakpoint only ADDS rules for wider viewports, never overrides them back down.

Three Cards, Two Widths

WIDE (ROW)



NARROW (COLUMN)



and, Comma, and not

Operator	Meaning
and	Combines conditions — ALL must be true
, (comma)	Works like OR — ANY one condition must be true
not	Inverts the entire media query (needs a media type)
orientation / prefers-color-scheme / print	Portrait vs. landscape / dark mode / printed page — same @media syntax

min-width and max-width, for Responsiveness

- **On an element:** min-width / max-width stop a box from shrinking or growing past a limit — a fluid container that is never too narrow or too wide.
- **In a media query:** @media (min-width: ...) and @media (max-width: ...) apply CSS conditionally based on the viewport's width — the mechanism behind every breakpoint.

```
.container { width: 100%; max-width: 1100px; min-width: 300px; margin: 0 auto; }

@media (min-width: 600px) { /* mobile-first: layer on rules as the screen grows */ }
```

- **Wide** — viewport $\geq 600\text{px}$

Chania

The Flight

The City

The Island

The Food

The City

Chania is the capital of the Chania region on the island of Crete.

The city can be divided in two parts, the old town and the modern city. The old town is situated next to the old harbour and is the matrix around which the whole urban area was developed.

Chania lies along the north west coast of the island Crete.

Facts:

- Chania is a city on the island of Crete
- Crete is a Greek island in the Mediterranean Sea

Resize the browser window to see the responsive effect.

- **Narrow** $< 600\text{px}$

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Resize the browser window to see the responsive effect.

► Try it yourself
on [W3Schools](https://www.w3schools.com)

The One Rule That Matters

```
img {  
  max-width: 100%;  
  height: auto;  
}
```

max-width: 100% keeps an image from ever growing wider than its container; height: auto preserves its aspect ratio.

The Same Image, Two Widths

WIDE

Container is 60% of the page width. The image's real file is 800×400px.



NARROW

Container is 60% of the page width. The image's real file is 800×400px.



Component-Based vs. Utility-First

BOOTSTRAP (COMPONENT-BASED)

- Ships pre-styled, ready-to-use components (.btn, .card, .navbar)
- Fast to get a decent-looking page up quickly
- Overriding built-in styles can fight the framework

TAILWIND (UTILITY-FIRST)

- Ships hundreds of tiny, single-purpose utility classes (flex, p-4)
- Nothing looks "designed" until you style it yourself
- Takes longer at first — many class names to learn

What Is Bootstrap, and Where Do You Get It?

- **The most widely used CSS framework** — originally built at Twitter; the current version, Bootstrap 5, needs no jQuery.
- **What it ships:** a 12-column responsive grid, dozens of pre-styled components (buttons, cards, navbars, modals, forms, alerts), plus a small JS bundle for interactive behaviour.
- **Where to get it:** getbootstrap.com — full documentation, live examples, and the exact CDN links to copy.
- **Also on npm** as the “bootstrap” package, for projects that already use a build tool such as Vite or Webpack.
- **Free and open source** (MIT licensed) — no signup, no cost, use it in any project.

Setting Up Bootstrap

CDN (FASTEST, ZERO BUILD STEP)

```
<link href="https://cdn.jsdelivr.net/npm/  
  bootstrap@5.3.3/dist/css/bootstrap.min.css"  
  rel="stylesheet">  
<script src="https://cdn.jsdelivr.net/npm/  
  bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js">  
</script>
```

- The CSS link goes in <head>; the JS bundle goes just before </body>.

NPM (FOR A BUNDLER-BASED PROJECT)

```
npm install bootstrap  
@import "bootstrap/dist/css/bootstrap.min.css";
```

- Import Bootstrap's JS in your entry file too, if you need dropdowns, modals, or other interactive components.
- Load Bootstrap's CSS before your own stylesheet, so your own rules win the cascade.

The 12-Column Grid

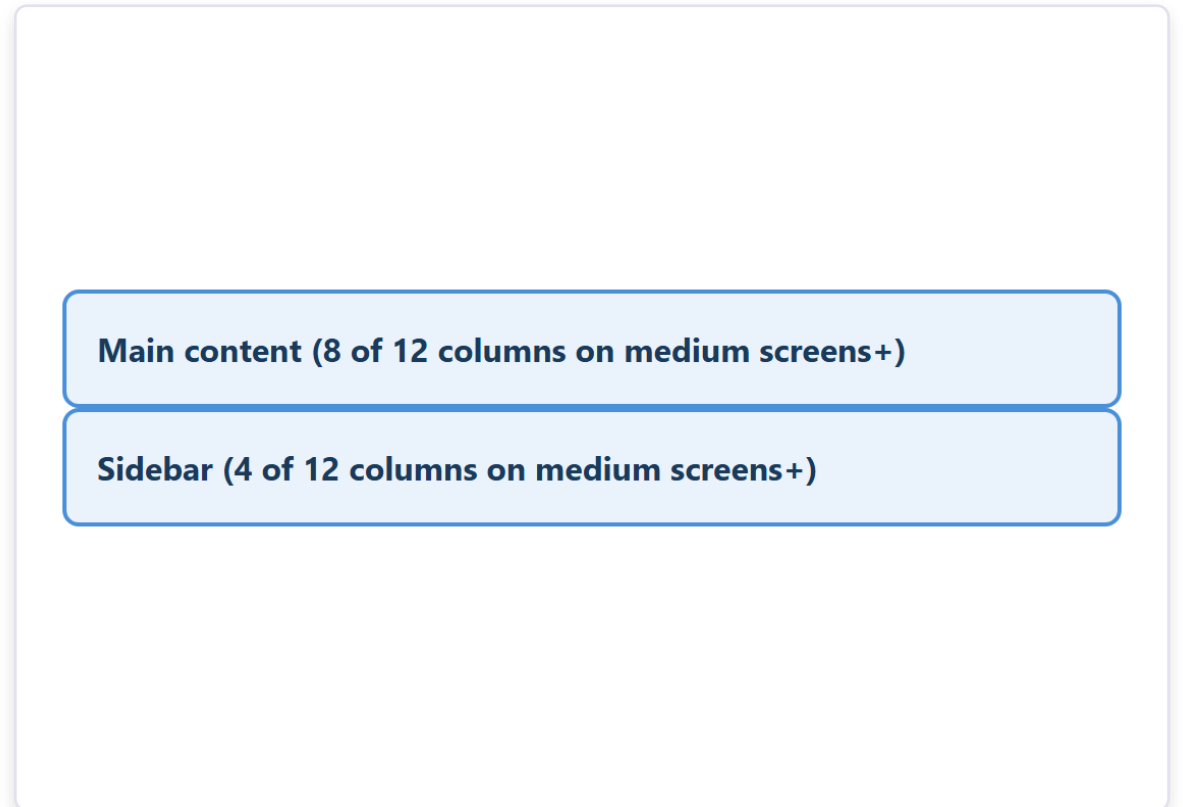
```
<div class="container">
  <div class="row">
    <div class="col-md-8">Main content</div>
    <div class="col-md-4">Sidebar</div>
  </div>
</div>
```

The Grid, Two Widths

WIDE (SIDE BY SIDE)



NARROW (STACKED)



Grid Modifier Classes

Class pattern	Effect
<code>.col (no number)</code>	Auto-layout: shares available width equally
<code>.col-{n}</code>	Exactly n of 12 columns, at all screen sizes
<code>.col-{bp}-{n}</code>	n columns from that breakpoint up (mobile-first)
<code>.offset-{bp}-{n}</code>	Pushes the column right by n empty columns
<code>.order-{n}</code>	Visual display order (0-5), independent of HTML order
<code>.row-cols-{n}</code>	Shorthand: every child column becomes 12/n wide
<code>g-{n} / g-{n}-xs / g-{n}-sm / g-{n}-md / g-{n}-lg / g-{n}-xl</code>	Grid wrapper around the Flexbox properties from Lecture 9 — <code>.row</code> is display: flex; <code>.col</code> is flex: 1; <code>.g</code> is flex-wrap: wrap; <code>.g-{n}</code> is flex-wrap: wrap; <code>.g-{n}-xs</code> is flex-wrap: wrap; <code>.g-{n}-sm</code> is flex-wrap: wrap; <code>.g-{n}-md</code> is flex-wrap: wrap; <code>.g-{n}-lg</code> is flex-wrap: wrap; <code>.g-{n}-xl</code> is flex-wrap: wrap

Auto-Layout Columns and Nesting

```
<div class="row">
  <div class="col">col</div>
  <div class="col">col</div>
  <div class="col">col</div>
</div>

<div class="row">
  <div class="col-8">
    col-8
    <div class="row">
      <div class="col-6">nested col-6</div>
      <div class="col-6">nested col-6</div>
    </div>
  </div>
  <div class="col-4">col-4</div>
</div>
```

Equal-width auto-layout columns (.col, no number)

col	col	col
-----	-----	-----

A row nested inside a .col

col-8	col-4
nested col-6	nested col-6

Offsetting and Reordering Columns

offset-md-4 centers a 4-wide column; order- rearranges visual order independent of HTML order.*

```
<div class="col-4 offset-md-4">col-4 offset-md-4</div>
```

```
<div class="order-3">First in HTML — order-3</div>
```

```
<div class="order-1">Second in HTML — order-1</div>
```

```
<div class="order-2">Third in HTML — order-2</div>
```

.col-4.offset-md-4 (centers a 4-wide column)

col-4 offset-md-4

.order-* (visual order independent of HTML order)

Second in HTML —
order-1

Third in HTML —
order-2

First in HTML —
order-3

Components, Rendered

Three lines of HTML, zero custom CSS.

```
<button class="btn btn-primary">Save Changes</button>

<div class="card"><div class="card-body">
  <h5 class="card-title">Card Title</h5>
</div></div>

<nav class="navbar navbar-expand-lg navbar-light bg-light">...</nav>
```

Save Changes

Card Title

Some quick example text for this card component.

Read more

MySite

Utilities, Rendered

```
<div class="d-flex justify-content-between p-3">  
  <!-- d-flex: display:flex; justify-content-between: space-between; p-3: padding -->  
</div>
```

Left item

Right item

What Is Tailwind CSS, and How Do You Set It Up?

- **Utility-first:** compose designs from small, single-purpose classes (flex, p-4, text-center) instead of writing custom CSS.
- **Where to get it:** tailwindcss.com — docs, plus an interactive Play CDN for instant prototyping.
- **Production setup:** npm install -D tailwindcss, then build with the Tailwind CLI/PostCSS so unused classes are purged and the shipped CSS stays small.

```
<!-- quick prototyping: one script tag, no build step -->
<script src="https://cdn.tailwindcss.com"></script>

# production build
npm install -D tailwindcss
npx tailwindcss -i input.css -o output.css --watch
```

A Fully Styled Button

```
<button class="bg-blue-600 hover:bg-blue-700 text-white font-semibold  
            px-4 py-2 rounded-lg shadow">  
  Save Changes  
</button>
```

The Button, Rendered



Save Changes

Responsive by Default

```
<div class="flex flex-col md:flex-row">  
  <!-- stacked by default, a row from md breakpoint up -->  
</div>
```

flex-col md:flex-row, Two Widths

WIDE (ROW)



NARROW (COLUMN)



Responsive Width

```
<div class="w-full lg:w-1/3">  
  <!-- full width by default, one-third width from the "lg" breakpoint up -->  
</div>
```

w-full lg:w-1/3, Two Widths

WIDE (1/3)

w-full by default, lg:w-1/3 from the "lg" breakpoint up

NARROW (FULL)

w-full by default, lg:w-1/3 from the "lg" breakpoint up

More Utility Categories

Category	Example classes
Flexbox	flex, justify-center, items-center, gap-4
Grid	grid, grid-cols-3, col-span-2
Sizing	w-1/2, h-screen, max-w-md
Borders	border, border-2, rounded-full
Typography	text-lg, font-bold, truncate
State variants	hover:, focus:, active:, disabled: (prefix any utility)

The spacing/color scale is numeric and consistent — $p-4 = 1\text{rem}$, blue-100 (near-white) through blue-900 (near-black).

Configuring Tailwind: theme.extend

```
module.exports = {  
  theme: {  
    extend: {  
      colors: { brand: '#ff6b35' }, // usable as bg-brand, text-brand...  
    },  
  },  
};
```

A Custom Color, Rendered

theme.extend adds to Tailwind's defaults, so bg-brand works with every state-variant prefix, just like a built-in color.

```
<button class="bg-brand hover:bg-orange-700 text-white font-semibold  
            px-4 py-2 rounded-lg">  
  Save Changes  
</button>
```

bg-brand (defined in tailwind.config.js theme.extend.colors)



Save Changes

Dark Mode

darkMode: 'class' in the config activates dark: utilities for any element inside a .dark ancestor.

```
<div class="bg-white dark:bg-gray-800 text-black dark:text-white">  
  Card content  
</div>
```

Default

Card content

.dark ancestor

Card content

The @apply Directive

```
.btn-primary {  
  @apply bg-blue-600 hover:bg-blue-700 text-white font-semibold  
    px-4 py-2 rounded-lg shadow;  
}  
  
<button class="btn-primary">Save Changes</button>
```

Produces the exact same button as writing all six utility classes by hand — @apply just names a repeated combination.

Bootstrap + Your Own Class

```
<button class="btn btn-primary my-cta-button">Get Started</button>
```

```
.my-cta-button { text-transform: uppercase; letter-spacing: 1px; }
```

Bootstrap + Custom, Rendered



GET STARTED

Tailwind + Your Own Class

```
<span class="text-blue-600 font-bold brand-underline">New!</span>
```

```
.brand-underline { text-decoration: underline wavy; text-underline-offset: 4px; }
```

Tailwind + Custom, Rendered

New!



What Is a CSS Preprocessor?

- A separate language that adds variables, nesting, and mixins on top of CSS, then COMPILES to plain CSS the browser reads
- Sass has two syntaxes: the indented .sass, and SCSS (.scss) — CSS-compatible braces/semicolons, by far the more common choice, including in Bootstrap's own source
- LESS solves the same problems with a different syntax (@var instead of \$var) — used by Bootstrap 3, before it switched to Sass

Getting Sass and LESS Running

SASS

- sass-lang.com — official site and docs.

```
npm install -D sass
sass styles.scss styles.css
sass --watch styles.scss:styles.css
```

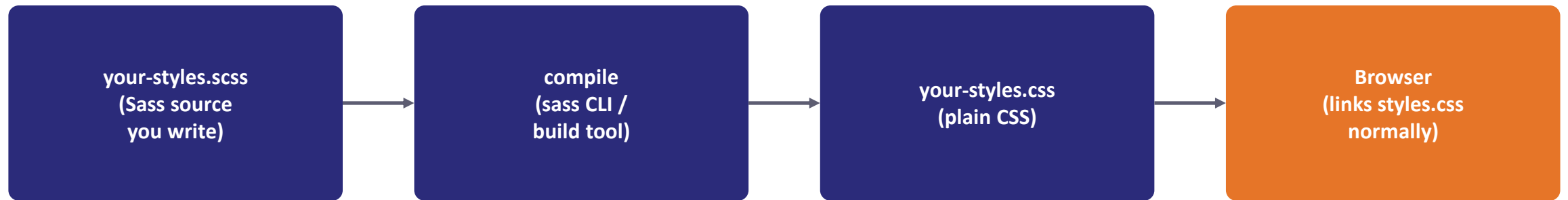
LESS

- lesscss.org — official site and docs.

```
npm install -D less
lessc styles.less styles.css
```

- A real project usually doesn't run these commands by hand — Vite or Webpack compiles Sass/LESS automatically through a loader or plugin.

Source vs. What the Browser Runs



The browser never sees Sass or LESS directly — it only ever loads the compiled, plain .css output, the same "source vs. what ships" split as modern JavaScript that gets compiled/bundled before it runs.

Sass Variables vs. CSS Custom Properties

	Sass (\$name)	CSS custom property (--name)
Resolved	At compile time — a fixed value	At runtime, in the browser
Changes after load?	No	Yes — JS or media queries can update it
Needs a build step?	Yes — must compile the .scss	No — works in a plain <style> tag

Variables, Nesting, and Mixins

```
$primary-color: #6c5ce7;

.card {
  border: 1px solid $primary-color;
  .title { color: $primary-color; }
  &:hover { box-shadow: 0 4px 10px rgba(0,0,0,0.2); }
}

@mixin flex-center { display: flex; justify-content: center; align-items: center; }
.banner { @include flex-center; background: $primary-color; }
```

Card Title

Some content styled with Sass.

Centered Banner

Partials and @use

```
// _variables.scss
$primary-color: #6c5ce7;

// main.scss
@use 'variables' as v;
.card { border: 1px solid v.$primary-color; }
```

A filename starting with _ is a partial — Sass pulls it into whatever file imports it, instead of compiling it separately.

Sass (SCSS) vs. LESS

Feature	Sass (SCSS)	LESS
Variable syntax	\$name	@name
Mixin definition	@mixin name { ... }	.name() { ... }
Mixin use	@include name;	.name();
Used by	Bootstrap 4 and 5, most new projects	Bootstrap 3, older codebases

Override Variables Before Importing

```
// custom.scss — override BEFORE importing Bootstrap's source
$primary: #ff6b35;
$border-radius: 1rem;
$grid-gutter-width: 3rem;

@import "bootstrap/scss/bootstrap";

// sass custom.scss custom.css
```

Every component reading these variables picks up the new values everywhere, automatically — no HTML changes at all.

Stock vs. Sass-Customized Bootstrap

STOCK BOOTSTRAP

Stock Bootstrap (default variables)

Save Changes

Card Title

Default radius and color.

Col	Col	Col
-----	-----	-----

SASS-CUSTOMIZED

Sass-customized Bootstrap (custom variables)

Save Changes

Card Title

Custom radius and color.

Col	Col	Col
-----	-----	-----

KEY TAKEAWAYS

Lecture 10 in Eight Points

1

Mobile-first: base styles for small screens, then add complexity with min-width media queries.

2

The viewport meta tag is required for media queries to work correctly on real phones.

3

Fluid units (% , rem, vw, vh) scale relative to something else, unlike fixed px.

4

Bootstrap's grid goes beyond col-md-*: auto-layout columns, nesting, offset-*, and order-* cover real layouts.

5

Bootstrap gives finished components; Tailwind gives composable utilities, configured via `tailwind.config.js` and `dark:`.

6

Sass/SCSS and LESS compile to plain CSS, adding variables (compile-time, unlike runtime CSS custom properties), nesting, and mixins.

7

Bootstrap's own source is Sass — overriding its variables before importing it recompiles the whole framework around your values.

8

Both frameworks expect your own custom CSS on top — they solve the common 80%, not the last 20% that makes a site unique.